Exercise 2

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In this exercise we are going to use pandas and seaborn to visualize data related to a kidney disease dataset. We begin by importing the libraries

In [77]:
```
!mamba install pandas
!mamba install seaborn
!mamba install xlrd
```

```
mambajs 0.19.13

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, pandas, seaborn, xlrd
Channels: emscripten-forge, conda-forge

Solving environment...
Solving took 0.7017000000029803 seconds
All requested packages already installed.
mambajs 0.19.13

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, pandas, seaborn, xlrd
Channels: emscripten-forge, conda-forge

Solving environment...
Solving took 0.32429999999701975 seconds
All requested packages already installed.
mambajs 0.19.13

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, pandas, seaborn, xlrd
Channels: emscripten-forge, conda-forge

Solving environment...
Solving took 0.2980999999940395 seconds
All requested packages already installed.
```

In [43]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import scipy as sp
```

a) Read the dataset using pandas' read_excel function. Use pandas.melt to transform the data from "wide" to "long" format, using class (indicating ckd for chronic kidney disease or notckd for its absence) as the identifier variable. (2P)

In [16]:
```
kidney_data = pd.read_excel("chronic_kidney_disease_numerical.xls", index_col =
kidney_data.head()
kidney_data = pd.melt(kidney_data, id_vars = "class")
```

In [76]:
```
kidney_data.head()
```

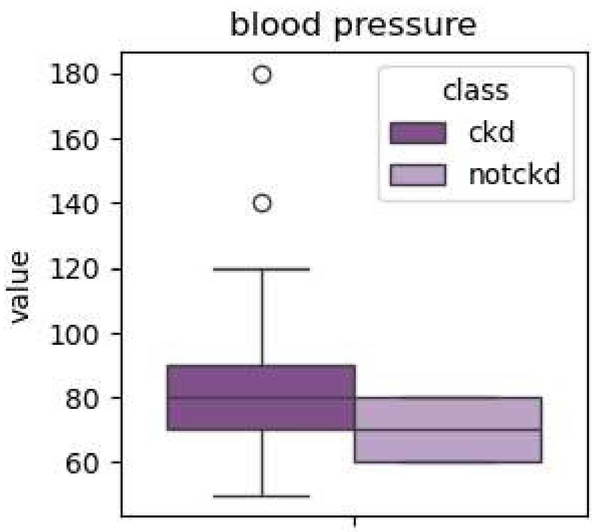
Out[76]:

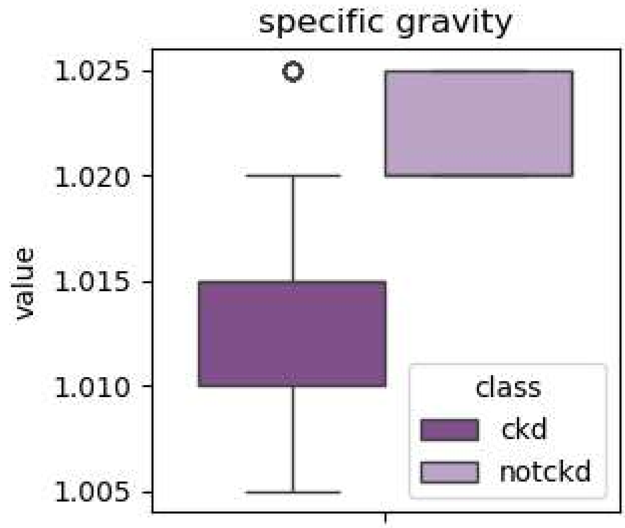
	class	variable	value
0	ckd	blood pressure	80.0
1	ckd	blood pressure	50.0
2	ckd	blood pressure	80.0
3	ckd	blood pressure	70.0
4	ckd	blood pressure	80.0

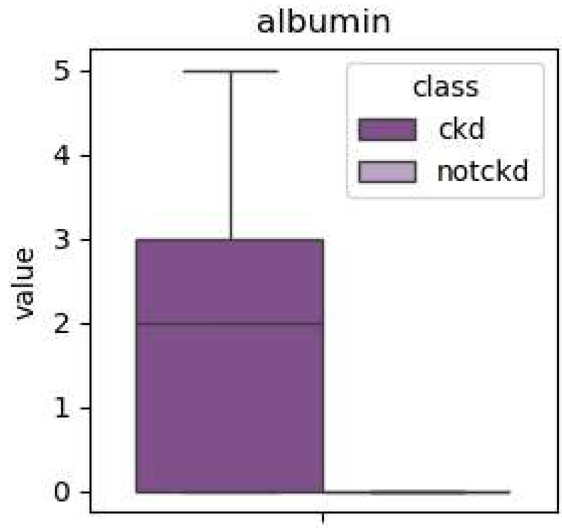
b) For each numerical attribute, such as age or blood pressure, create two boxplots side-by-side. One should show the attribute's distribution among patients suffering from chronic kidney disease, the other one from patients who do not suffer from the disease. Hint: Due to the different numerical ranges, you will have to disable sharing of y axes between plots of different attributes. (4P)

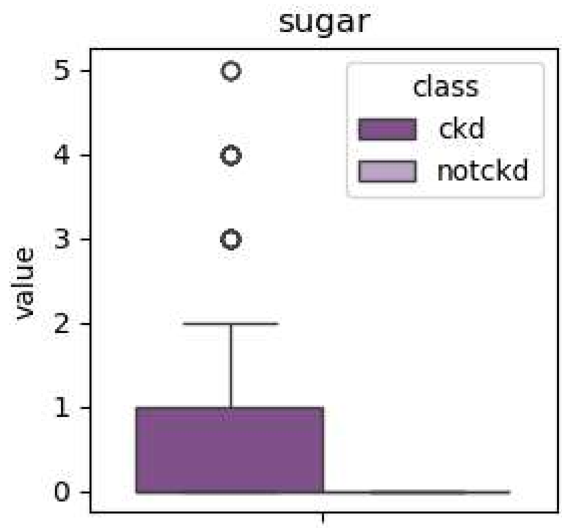
In [74]:

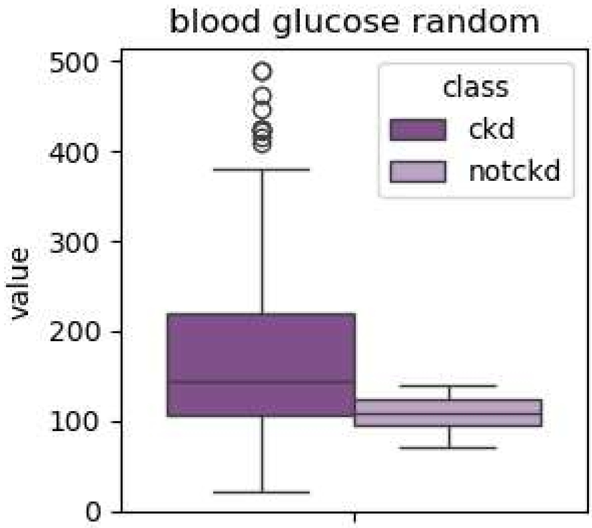
```
#fig, ax = plt.subplots(4, len(kidney_data["variable"].unique())//4 + 1)
sns.set_palette("PRGn")

for i in range(len(kidney_data["variable"].unique())):
    plt.figure(figsize = (3, 3))
    sns.boxplot(data = kidney_data[kidney_data["variable"] == kidney_data["varia
    plt.title(kidney_data["variable"].unique()[i])
    plt.show()
```

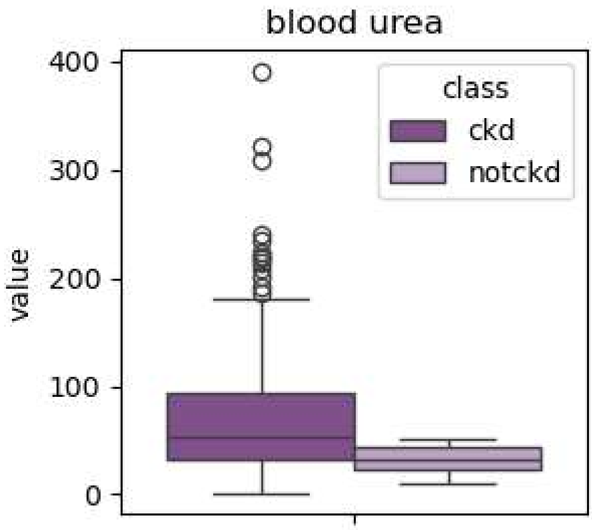


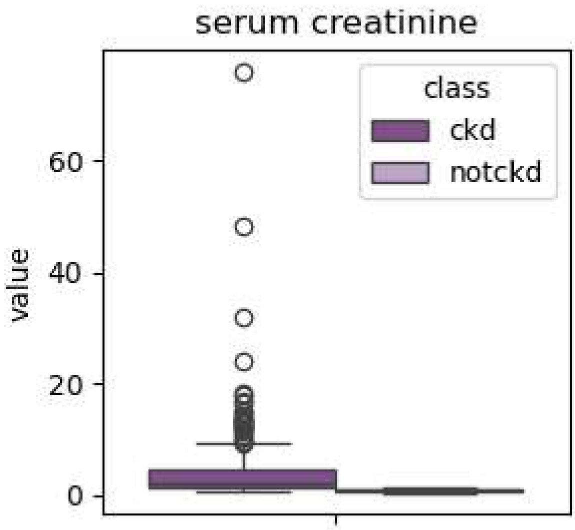

specific gravity
value
1.025
1.020
1.015
1.010
1.005
class
ckd
notckd

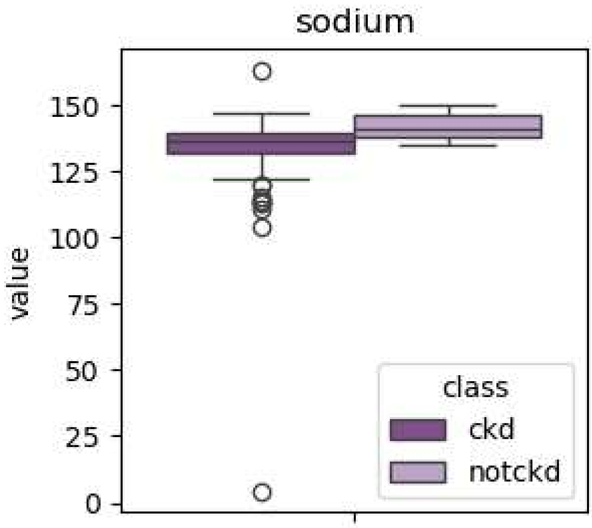

albumin
value
5
4
3
2
1
0
class
ckd
notckd

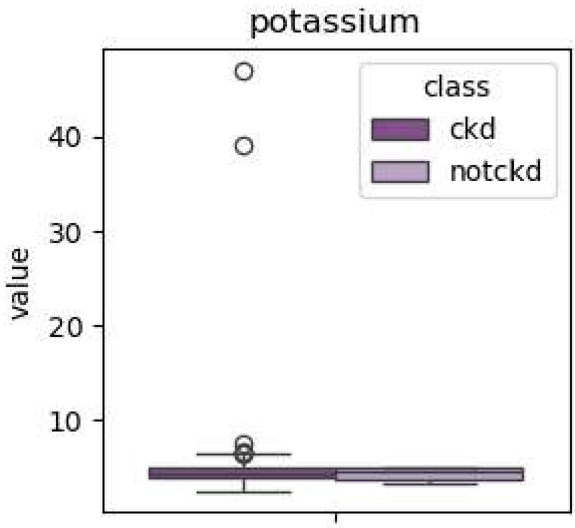

sugar
value
5
4
3
2
1
0
class
ckd
notckd

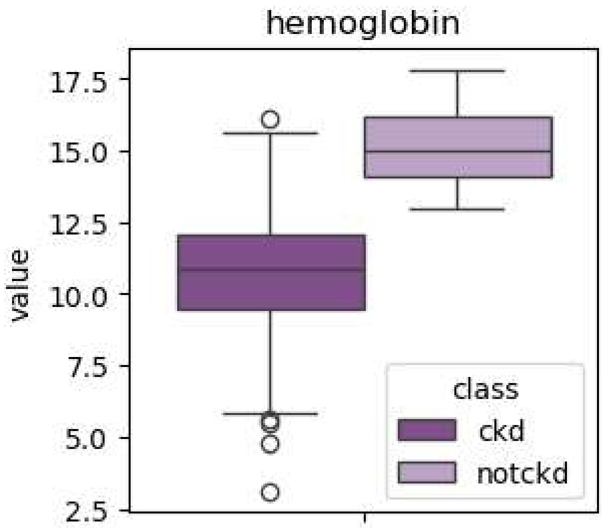

blood glucose random
value
500
400
300
200
100
0
class
ckd
notckd

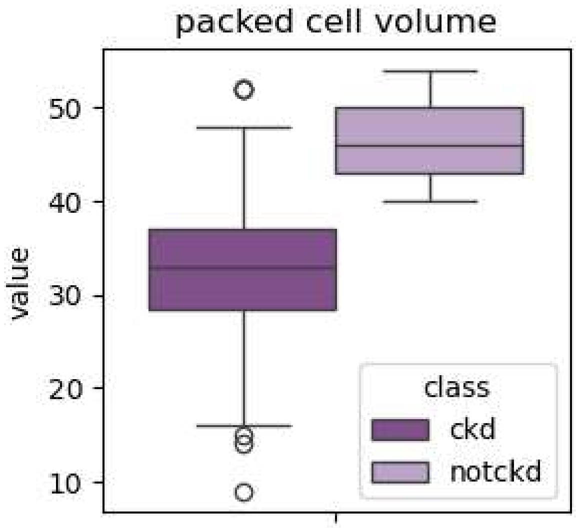

blood urea
value
400
300
200
100
0
class
ckd
notckd

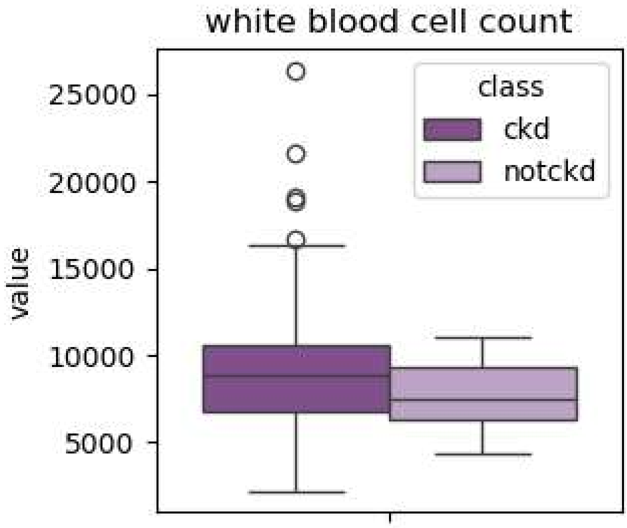

serum creatinine
value
60
40
20
0
class
ckd
notckd

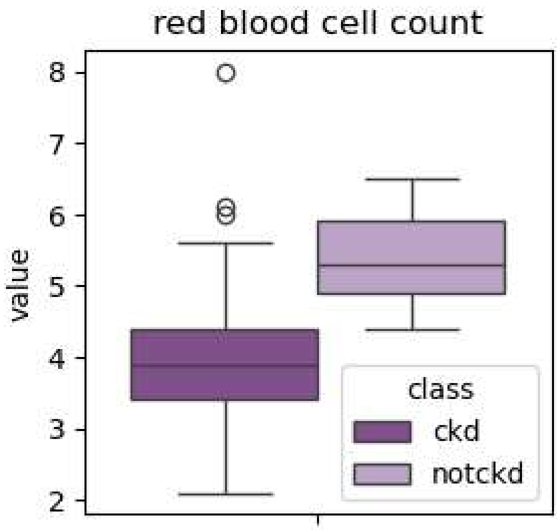

sodium
value
150
125
100
75
50
25
0
class
ckd
notckd


potassium
value
40
30
20
10
class
ckd
notckd


hemoglobin
value
17.5
15.0
12.5
10.0
7.5
5.0
2.5
class
ckd
notckd







c) Based on viewing the plots, name an attribute that appears to be highly indicative of chronic kidney disease, and one that seems to be mostly unrelated to it. (1P)

Observing the plots, we see that the attributes *albumin* and *sugar* are highly sensitive to kidney disease (in fact, every single person that doesn't suffer from it observes a 0 in both categories). Meanwhile, *potassium* levels seem to not be affected by this disease.